10 · Project sheet · Ayo Goes to the Shop

Robot / Maker Lab practical · Nursery 1 · Term 3 · Week 2

1. At a glance

Time	40 minutes (fits the lesson; 15 min if used on its own)
Group size	Groups of 4–6
Learning goal	I can break a job into small steps.
Coding idea	Decomposition: plan the small steps first, put them in order, then act.
Week link	https://rhinoxscientifica.com/q/n1-t3-w2

2. Materials needed

Item	Per	Low-cost swap
Shop step cards + FIRST–NEXT–LAST strip	group	Draw them
Box shop + paper bread	class	A chair as the shop
A mat to roll	group	A towel or cloth
Plastic jug, bowl, large card counters	group	Cup and box
Cardboard Ayo	class	A toy
Step Star stickers	child	Washable marker star on the hand

3. Safety and care

- Counters bigger than a toilet-roll opening; count in and out.
- Plastic jug and bowl only.
- Always cross the road with a grown-up (in acting and in real life).
- Walk, don’t run, to the shop.

4. Before the lesson

1. Set up the box shop, mat and jug station.
2. Print and cut the step cards and strips.
3. Count the counters.

5. Step-by-step guide

Step	Teacher says	Children do
1	“Ayo needs bread. That is a big job!”	Listen; arms wide for “big job”
2	“Plan first: tell me the small steps.”	Say the steps in own words
3	“Now choose the cards and put them in order.”	Lay leave home, buy bread, go home
4	“Act the plan! Cross with a grown-up.”	Act the 3 steps
5	“Break another job: roll the mat.”	Say steps, lay cards, roll the mat

Step	Teacher says	Children do
6	“Stretch: what step could we add?”	Add “pay” as step 4

6. Explore it

Question	Is a big job easier in small steps?
Predict	“Can Ayo do the whole shop trip in one go?”
Try	Do it in small steps, one card at a time.
Observe	“Did each small step help?”
Explain	Small steps make a big job easy to plan, do and check.

7. Talk about it

1a. What is the big job?

1b. What are the small steps?

1c. Which step comes first?

2a. What are the small steps to roll a mat?

2b. What happens if we skip a step?

2c. What step could we add?

8. Support and stretch

Support	Start with 2 steps, then add the middle one; act it together first.
Stretch (optional)	Break a bigger job into 4 steps. Never assessed.

9. How we check

Observation only, not marked.

Look for	Secure	With prompts	Not yet
Says the small steps before choosing cards	☐	☐	☐
Orders the 3 step cards	☐	☐	☐
Acts the plan	☐	☐	☐

10. Take-home and parent note

> **Dear parent or carer**
> Today we **broke a big job into small steps**: leave home, buy bread, go home.
> At home: before a job, ask “What are the small steps?”

11. Show and share

- Groups act their shop trip or mat-rolling for the class, saying each step.

- Everyone says: “Break it small!”
- Every child who said the steps and ordered them gets a **Step Star sticker**.



The Step Star is a weekly sticker, not a badge. The term badge “Step Thinker · Term 3” is awarded in Week 13.